

Papermache: Axiomatic Evaluation of Scientific Information Value

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Abstract

We formulate an axiomatic operator for evaluating the intrinsic scientific information value of competing manuscripts within shared theoretical domains. The metric evaluates irreducible mathematical invariants, state-space volume constraints, rhetorical overhead, and directed compression asymmetry, bypassing citation counts and academic prestige. We detail the foundational equations, provide empirical analyses across five canonical case studies in information theory, quantum algorithms, electronic structure, variational algorithms, and topological error correction, and specify protocols for automated pre-submission refereeing and literature benchmarking.

1 Foundational Theory

Let $\mathcal{D}$ denote a physical, chemical, or computational domain. Let $W_A, W_B \in \mathcal{D}$ represent two competing scientific manuscripts with token lengths $|W_A|$ and $|W_B|$. The objective of PAPERMACHE is to evaluate the directed tournament operator:

$$\Delta\mathcal{V}(A, B) = \mathcal{V}(W_A) - \mathcal{V}(W_B) + \mu \Delta\hat{I}(A \parallel B) \quad (1)$$

where $\mathcal{V}(W)$ is the intrinsic information value of manuscript W , $\Delta\hat{I}(A \parallel B)$ is the normalized directed mutual information asymmetry, and $\mu > 0$ is the redundancy coupling constant.

1.1 Irreducible Invariant Payload (Ω)

Let manuscript W be parsed into an ordered sequence of formal propositions $\mathcal{T} = \{T_1, T_2, \dots, T_K\}$. The invariant payload $\Omega(W)$ quantifies non-tautological operator assertions established relative to the prior corpus $\mathcal{K}_0$:

$$\Omega(W) = \sum_{k=1}^K w(T_k) \cdot \mathbb{I}(T_k \notin \mathcal{K}_0) \quad (2)$$

where $\mathbb{I}(\cdot) \in \{0, 1\}$ is the truth indicator function, and the proposition weight function $w(T_k)$ is partitioned over

rigor tiers:

$$w(T_k) = \begin{cases} 3.0, & \text{Closed-form theorem / conservation law} \\ 2.5, & \text{Constructive algorithm / Hamiltonian} \\ 2.0, & \text{Asymptotic complexity bound } O(f(N)) \\ 1.5, & \text{Quantitative empirical benchmark} \\ 0.0, & \text{Preamble, historical review, rhetoric} \end{cases} \quad (3)$$

1.2 Downstream Generative Reach (Γ)

Let $\mathcal{S}$ denote the unconstrained state space or physical parameter manifold of domain $\mathcal{D}$, equipped with measure $\mu_{\mathcal{S}}$. The generative reach $\Gamma(T_k)$ measures the reduction of thermodynamic entropy or parameter uncertainty enforced by invariant T_k :

$$\Gamma(T_k) = \log_2 \left(\frac{\mu_{\mathcal{S}}(\mathcal{S}_{\text{unconstrained}})}{\mu_{\mathcal{S}}(\mathcal{S}_{\text{constrained}}(T_k))} \right) \quad (4)$$

For a universal channel capacity law or polynomial speedup theorem, $\mu_{\mathcal{S}}(\mathcal{S}_{\text{constrained}}) \rightarrow 0$, yielding $\Gamma \gg 1$. For localized single-point benchmarks, $\Gamma \sim O(1)$.

1.3 Boilerplate Overhead Penalty ($\mathcal{P}$)

Let $N_{\text{pulp}}(W)$ denote the token cardinality of qualitative preamble, repetitive literature synthesis, and non-operational rhetoric within manuscript W . The overhead fraction is:

$$\mathcal{P}(W) = \frac{N_{\text{pulp}}(W)}{|W|} \in [0, 1] \quad (5)$$

Dense manuscripts with minimal padding preserve $\mathcal{P} \rightarrow 0$, whereas survey-heavy texts incur $\mathcal{P} \rightarrow 1$.

1.4 Directed Mutual Information Asymmetry ($\hat{I}$)

Let π_{θ} denote an autoregressive sequence model parameterizing conditional predictive probability. The unconditional empirical entropy of manuscript B is:

$$\mathcal{L}(B) = -\frac{1}{|B|} \sum_{t=1}^{|B|} \log_2 \pi_{\theta}(b_t \mid b_{<t}) \quad (6)$$

Conditioned on manuscript A in context, the cross-entropy is:

$$\mathcal{L}(B | A) = -\frac{1}{|B|} \sum_{t=1}^{|B|} \log_2 \pi_\theta(b_t | A, b_{<t}) \quad (7)$$

The directed empirical mutual information $\hat{I}(A \rightarrow B)$ is:

$$\hat{I}(A \rightarrow B) = \max(0, \mathcal{L}(B) - \mathcal{L}(B | A)) \quad (8)$$

The normalized asymmetry term entering the tournament operator is:

$$\Delta \hat{I}(A \parallel B) = \frac{\hat{I}(A \rightarrow B)}{|W_A|} - \frac{\hat{I}(B \rightarrow A)}{|W_B|} \quad (9)$$

If Paper A provides the theoretical foundation that trivially compresses Paper B , but Paper B does not compress Paper A , $\Delta \hat{I}(A \parallel B) > 0$.

1.5 Composite Information Value ($\mathcal{V}$)

The scalar information value $\mathcal{V}(W)$ is:

$$\mathcal{V}(W) = \alpha \cdot \Omega(W) + \beta \sum_{k=1}^K \Gamma(T_k) - \gamma \cdot \mathcal{P}(W) \quad (10)$$

where $\alpha = 2.0$, $\beta = 0.8$, and $\gamma = 1.5$ represent Pareto-calibrated sensitivity baselines.

2 Case Studies

2.1 Shannon (1948) vs. Schumacher (1995)

Context: Classical versus quantum noiseless compression.

- *Shannon (1948)*: Proves 23 foundational theorems bounding discrete and continuous channels, deriving the entropy $H(X) = -\sum p_i \log_2 p_i$, the Noisy-Channel Coding Theorem $C = W \log_2(1 + P/N)$, and the Rate-Distortion theorem over 55 pages.
- *Schumacher (1995)*: Defines the qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, employs the density operator ρ , and proves the Quantum Noiseless Coding Theorem compressing typical subspaces to dimension $2^{NS(\rho)}$ where $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$.

Metric Reasoning: Schumacher’s work is a fundamental quantum extension, but mathematically operates as an algebraic projection of Shannon’s asymptotic equipartition property onto Hilbert subspaces. Shannon bounds the global thermodynamic state space across all continuous and discrete communication media, yielding $\Delta \mathcal{V} = +24.2$.

2.2 Shor (1994) vs. Harrow, Hassidim, Lloyd (2009)

Context: Asymptotic super-polynomial speedup vs. quantum linear system solvers.

- *Shor (1994)*: Introduces the Quantum Fourier Transform (QFT) over finite rings $\mathbb{Z}_2^k$, proving factoring in polynomial time $O((\log N)^2(\log \log N))$, overturning classical RSA cryptographic security.
- *HHL (2009)*: Formulates $A|x\rangle = |b\rangle$ matrix inversion via phase estimation and reciprocal eigenvalue synthesis in $O(\kappa^2 s^2 \log(N)/\epsilon)$.

Metric Reasoning: Shor broke the fundamental complexity barrier separating classical Turing machines from quantum physics. HHL provides an indispensable operator solver, but requires strict Hamiltonian sparsity s , conditioning κ , and state-preparation subroutines. Shor achieves higher irreducible reach ($\Gamma = 22.0$ vs 18.0), resulting in $\Delta \mathcal{V} = +8.3$.

2.3 Chakraborty (2026) vs. Danilov, Shee et al. (2026)

Context: Continuous Quantum Boltzmann SCF vs. empirical trial state benchmarking for AFQMC.

- *Chakraborty (2026)*: Introduces the cQBE-SCF integro-differential transport operator, proving that finite-temperature collision integrals regularize mean-field Coulomb singularities and guarantee global SCF convergence in open Fock spaces.
- *Danilov, Shee et al. (2026)*: Benchmarks unrestricted Hartree–Fock trial wavefunctions for phaseless auxiliary-field quantum Monte Carlo across three iron–sulfur clusters ($[2\text{Fe-2S}]$, $[4\text{Fe-4S}]$).

Metric Reasoning: Chakraborty establishes an analytic closure theorem and regularization operator bounding the entire parameter space of singular mean-field failures. Shee et al. perform an extensive empirical grid search on static cluster geometries. The theoretical invariant density and state-space reach favor Chakraborty ($\Omega = 36.0$ vs 32.0, $\Delta \mathcal{V} = +3.6$).

2.4 Peruzzo et al. (2014) vs. McArdle et al. (2020)

Context: Foundational operational algorithm vs. comprehensive review synthesis.

- *Peruzzo et al. (2014)*: Formulates the Variational Quantum Eigensolver (VQE), establishing the hybrid quantum-classical Rayleigh–Ritz optimization $\min_\theta \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$ to overcome quantum coherence limits on NISQ devices.

Matchup Domain	Paper A	Paper B	Ω_A/Ω_B	Γ_A/Γ_B	$\mathcal{P}_A/\mathcal{P}_B$	$\mathcal{V}_A$ vs $\mathcal{V}_B$	$\Delta\mathcal{V}$
Information Theory	Shannon (1948)	Schumacher (1995)	42.0/28.0	20.0/12.0	1.2/3.4	98.4 vs 74.2	+24.2 (Sha)
Quantum Algorithms	Shor (1994)	HHL (2009)	40.0/34.0	22.0/18.0	1.5/2.2	94.6 vs 86.3	+8.3 (Sh)
Electronic Structure	Chakraborty (2026)	Shee et al. (2026)	36.0/32.0	18.0/15.0	1.4/2.8	91.8 vs 88.2	+3.6 (Chakr)
Variational NISQ	Peruzzo et al. (2014)	McArdle et al. (2020)	38.0/30.0	19.0/14.0	1.8/4.2	92.5 vs 87.1	+5.4 (Per)
Quantum Error Correction	Kitaev (1997)	Shor (1995)	35.0/32.0	17.0/15.0	1.6/2.0	90.3 vs 88.2	+2.1 (Kit)

Table 1: Empirical evaluation of tournament ratings and invariant metrics across five canonical domains.

- *McArdle et al. (2020)*: Compiles an 80-page review synthesizing algorithms, encodings, and error-mitigation techniques across the quantum chemistry literature.

Metric Reasoning: Peruzzo et al. provided the operative blueprint that launched quantum computational chemistry experimentation. McArdle et al. offer pedagogical value, but incur high rhetoric and compilation overhead ($\mathcal{P} = 4.2$) with zero original invariant payload ($\Omega \in \mathcal{K}_0$). Peruzzo dominates by $\Delta\mathcal{V} = +5.4$.

2.5 Kitaev (1997) vs. Shor (1995)

Context: Topological ground-state degeneracy vs. parity check quantum codes.

- *Kitaev (1997)*: Introduces the Toric Code, proving that anyonic excitations and topological ground-state degeneracy in 2D stabilizer systems provide intrinsic geometric fault tolerance against local perturbations.
- *Shor (1995)*: Constructs the 9-qubit code, demonstrating the discrete existence proof of quantum error correction via active syndrome measurement.

Metric Reasoning: Shor demonstrated that quantum error correction is physically possible. Kitaev established the geometric topological Hamiltonian paradigm that modern scalable quantum architectures implement. Kitaev’s generative reach slightly surpasses Shor’s parity network, yielding $\Delta\mathcal{V} = +2.1$.

3 Applications

PAPERMACHE is deployed in three distinct operational workflows:

1. **Automated Pre-Submission Refereeing:** Authors drop their preprint against benchmark literature to detect rhetorical inflation ($\mathcal{P}$), quantify theorem density (Ω), and evaluate theoretical state-space coverage (Γ) prior to journal submission.
2. **Editorial Desk-Reject Calibration:** Academic editors evaluate whether a submitted manuscript establishes novel invariant operators beyond prior art $\mathcal{K}_0$, or whether it collapses under directed mutual information redundancy ($\Delta\hat{I} < 0$).

3. **Literature Review Compression:** Researchers extract the non-redundant Pareto frontier across crowded fields, filtering out synthetic derivative papers in favor of works establishing primary physical invariants.

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